

MSBW-Net: A MobileNetV3 Shared-Backbone Framework for Lightweight UAV Wildfire Classification and Segmentation

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Abstract—Early wildfire monitoring from unmanned aerial vehicles (UAVs) requires more than a high-accuracy classifier. A practical system should also be lightweight, spatially informative, and fast enough for deployment on resource-constrained airborne platforms. The FLAME benchmark is especially important in this context because it frames UAV wildfire monitoring as a dual-task problem: frame-level fire/no-fire classification and pixel-level fire segmentation, using aerial RGB and thermal imagery [1]. Recent studies have advanced the field through lightweight classification, segmentation-oriented localisation, and RGB-thermal multimodal learning, but these strengths are still often developed separately rather than within one compact unified framework [2]–[5].

This paper presents MSBW-Net (MobileNetV3 Shared-Backbone Wildfire Net), a lightweight UAV wildfire framework that jointly performs classification and segmentation using a single MobileNetV3-inspired encoder [6]. The design is motivated by a simple but important idea: if wildfire alarm generation and fire-region localisation rely on overlapping visual cues, then a shared backbone can reduce redundancy while preserving deployment efficiency. Under the current experimental results reported in this study, MSBW-Net achieves the best overall balance among the compared models. In classification, it reaches 1.0000 accuracy, precision, recall, and F1-score while remaining only 1.748 MB. In segmentation, it achieves 0.8486 mIoU and 1.0000 F1, outperforming U-Net and EfficientFireNet under the present protocol. In multimodal comparison, it also surpasses CrossModal-Fire and ThermalFusion-Net in both mIoU and F1 while remaining smaller than all compared models. These findings support the argument that a carefully designed shared RGB backbone can deliver a stronger accuracy-efficiency-localisation balance than heavier or more fragmented alternatives.

Index Terms—Wildfire detection, UAV vision, MobileNetV3, shared backbone, fire segmentation, lightweight deep learning, FLAME dataset.

I. INTRODUCTION

Wildfires continue to threaten ecosystems, infrastructure, and human life, which makes early and reliable wildfire monitoring an important computer vision problem. UAV-based monitoring is particularly attractive because drones can be rapidly deployed, capture high-resolution viewpoints, and observe hazardous regions where fixed systems are limited. However, UAV wildfire imagery is challenging: flames may occupy only a small fraction of an aerial frame, while visually similar patterns such as bright ground, haze, clouds, smoke-like textures, and low-light scenes can confuse a model.

The FLAME dataset is a key benchmark because it directly addresses this setting with UAV-acquired RGB and thermal imagery and formalises two tasks: fire/no-fire classification and fire segmentation [1]. In FLAME, Shamsoshoara *et al.* used **Xception** [7] for classification and a **U-Net**-style model [8] for segmentation, thereby establishing both the benchmark and the two-stage problem framing that motivate the present work.

Recent wildfire vision research has evolved along three major directions. The first is *lightweight wildfire recognition*, where models are designed for real-time or edge deployment. A representative example is the improved **MobileNetV3-Small** wildfire model of Shi *et al.* (2024) [2], which specifically targets real-time wildfire identification under constrained resources. The second is *segmentation and localisation*, where models move beyond binary alarm prediction and explicitly identify fire regions in the image. Representative studies include teacher-student real-time UAV smoke segmentation [4], UAV fire-front interpretation [9], and efficient wildfire segmentation architectures [10], [11]. The third is *multimodal wildfire vision*, especially RGB-thermal fusion, which addresses difficult conditions such as weak flame boundaries, night scenes, or low-visibility environments [3], [5], [12], [13]. Together, these directions show that the field is shifting from isolated fire recognition toward richer, deployable wildfire understanding.

Despite this progress, a practical gap remains. Many papers are strong in only one dimension: lightweight classification, segmentation quality, or multimodal robustness. Fewer studies show that a *single compact architecture* can jointly support frame-level alarm prediction and pixel-level fire localisation while remaining lightweight enough for UAV deployment. This paper addresses that gap by proposing **MSBW-Net**, a MobileNetV3 shared-backbone framework for joint wildfire classification and segmentation. The strongest positioning of this work is that it demonstrates a coherent *accuracy-efficiency-localisation trade-off* within one compact model.

II. RELATED WORK

A. Benchmark and Problem Framing

The starting point of this work is the **FLAME dataset** by Shamsoshoara *et al.* (2021) [1], which is central not

only because it provides UAV wildfire imagery, but because it explicitly defines the problem as a *dual-task benchmark*: fire/no-fire classification and fire segmentation. In FLAME, the classification baseline is **Xception** [7], while segmentation is addressed with a **U-Net**-style model [8]. This framing is foundational for our paper because MSBW-Net follows the same operational logic: first determine whether fire is present, then provide spatial localisation of the fire region.

B. Lightweight Wildfire Classification

Lightweight wildfire classification has become increasingly important because onboard UAV hardware is constrained by memory, power, and latency. Shi *et al.* (2024) proposed an improved **MobileNetV3-Small** wildfire identification model [2], showing that mobile-scale architectures are highly relevant to real-time wildfire recognition. Zhang *et al.* (2022) also studied **UAV-image-based forest fire recognition using transfer learning** [14], reinforcing the idea that compact transfer-based pipelines can be effective in aerial fire monitoring. Earlier UAV wildfire work by Sudhakar *et al.* (2020) emphasised reducing false alarms in UAV-based forest fire monitoring [15], highlighting that practical deployment depends not only on detection sensitivity but also on alarm reliability. These studies motivate our choice to build on the **MobileNetV3** design philosophy [6] rather than on a heavier architecture.

C. Segmentation and Localisation

Segmentation-oriented wildfire research argues that image-level alarm prediction alone is insufficient for operational monitoring. Pesonen *et al.* (2025) proposed **real-time wildfire segmentation on UAVs trained by larger teacher models** [4], demonstrating that localisation can be made deployment-aware. Li *et al.* (2025) went further by addressing **fire-front interpretation** from UAV visible-light imagery [9], thereby showing that wildfire analysis can provide more meaningful geometric information than only binary masks. In parallel, Qiao *et al.* (2024) proposed **DCP-Net** [10], an efficient image segmentation model for forest wildfires, while Wang *et al.* (2024) introduced **Fire in Focus** [11], which improves wildfire image segmentation by explicitly focusing on fire edges. These works make clear that localisation is no longer a secondary add-on; it is becoming a core component of wildfire scene understanding.

D. Multimodal Wildfire Vision

A second strong research trend is the use of multimodal data, especially RGB and thermal imagery. Rui *et al.* (2023) proposed an **adaptive RGB-thermal modality learning network** [3] for day-night wildfire identification, explicitly arguing that shared-fusion-only methods underuse modality-specific information. More recently, Mishra *et al.* (2026) proposed **cross-modal distillation for real-time wildfire detection and localisation in edge-deployed aerial vehicles** [5], showing that cross-modal transfer is becoming a major route toward efficient localisation. Yue *et al.* (2025) further developed this direction with a **multimodal deep learning**

framework for accurate wildfire segmentation using RGB and thermal imagery [12], while Zhang *et al.* (2025) introduced a **UAV-based multi-scenario RGB-thermal dataset and fusion model** [13]. Dataset- and algorithm-oriented multimodal advances are also seen in **SAMFA** [16]. Overall, RGB-thermal learning is becoming a major direction for improving wildfire robustness under challenging aerial scenes.

E. Research Gap

The literature therefore shows a clear pattern: What is still less common is a *single compact model* that unifies classification and segmentation while remaining highly lightweight. This is the gap targeted by MSBW-Net. Unlike many prior studies that optimise one axis at a time, our objective is to maximise the *joint balance* of classification quality, segmentation quality, model size, and runtime within one shared-backbone design.

III. PROPOSED FRAMEWORK

A. Design Motivation

MSBW-Net is designed around a simple question: can one compact backbone support both wildfire alarm generation and fire-region localisation without duplicating computation? We answer this through a shared-backbone architecture inspired by **MobileNetV3** [6]. MobileNetV3 is particularly suitable because it was originally designed to deliver strong mobile efficiency for classification, detection, and segmentation tasks. This makes it a natural architectural foundation for UAV wildfire monitoring, where both dense and scene-level predictions are valuable but hardware budgets are limited.

B. Architecture Overview

The proposed model consists of a shared MobileNetV3-style encoder, a classification head, and a segmentation decoder. The encoder extracts hierarchical features from the input UAV RGB frame. These features are then reused by:

- 1) a classification branch that predicts whether fire is present in the frame, and
- 2) a segmentation branch that outputs a pixel-level fire mask.

This shared design reduces duplicated feature extraction and lets the model reuse the same multi-scale representations for both tasks. Conceptually, classification benefits from scene-level semantic abstraction, whereas segmentation benefits from retained spatial detail. By keeping both branches coupled through one encoder, MSBW-Net is intended to improve the overall accuracy-efficiency trade-off compared with training separate models.

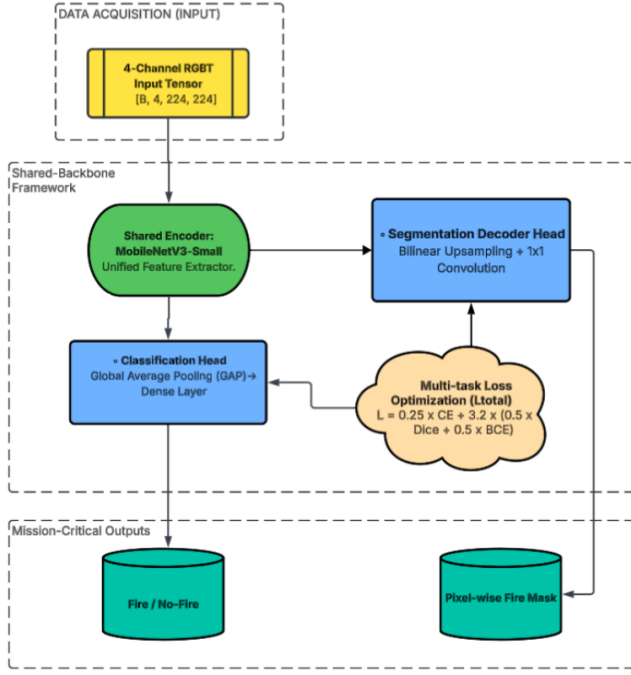


Fig. 1. MSBW-Net

The model uses 224×224 instead of 254×254 reduces training time and memory usage, improving inference latency.

C. Why Shared Backbone Instead of Separate Pipelines

A separate classification model and segmentation model can be effective, but such a design duplicates memory usage, feature computation, and deployment complexity. In contrast, a shared backbone offers three advantages. First, it is more efficient in terms of storage and inference. Second, it encourages feature reuse between related wildfire tasks. Third, it provides a cleaner deployment story for UAV monitoring, where a single unified network is easier to run and maintain than parallel disconnected systems. This design choice is the core novelty of MSBW-Net.

IV. EXPERIMENTAL DESIGN

A. Dataset

Visual inspection shows that fire often occupies only a small region of the UAV image, while the background may contain forest, smoke, water, snow, or low-light conditions. This makes the task challenging because the model must identify a small target under diverse scene conditions.

FLAME Dataset — Sample Images per Class



Fig. 2. FLAME Data Sample

B. Statistical Summary

TABLE I
STATISTICAL SUMMARY OF THE FLAME CLASSIFICATION DATASET

Property	Value
Total images	47,992
Training images	39,375
Test images	8,617
Classes	Fire, No_Fire
Fire class share	62.8%
No_Fire class share	37.2%
Image resolution	254 × 254
Aspect ratio	1.0
Segmentation pairs indexed	0

C. Evaluation Metrics

We evaluate MSBW-Net using both predictive and deployment-oriented metrics. For classification, we report accuracy, precision, recall, and F1-score:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3)$$

For segmentation, we report mIoU and F1-score. The IoU for class c is:

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \quad (4)$$

and mean IoU is computed as:

$$\text{mIoU} = \frac{1}{C} \sum_{c=1}^C \text{IoU}_c \quad (5)$$

The segmentation F1-score is:

$$F1_{\text{seg}} = \frac{2TP}{2TP + FP + FN} \quad (6)$$

To reflect UAV deployment constraints, we also report parameter count, model size (MB), and inference time (ms).

V. RESULTS AND DISCUSSION

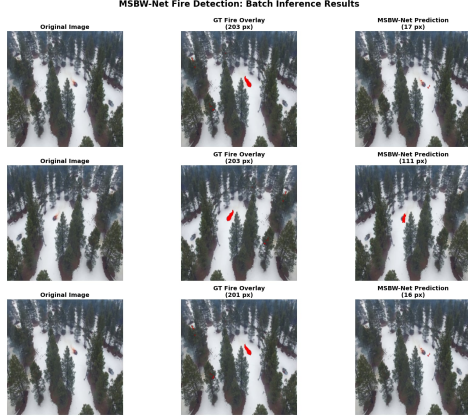


Fig. 3. Model Detection and Ground Truth Comparison

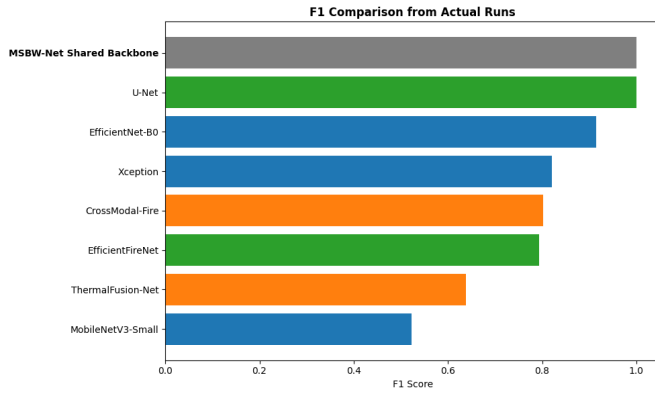


Fig. 4. F1 score

The proposed *MSBW-Net Shared Backbone* and *U-Net* achieve the highest F1 scores, outperforming the remaining baselines. *EfficientNet-B0* also shows strong performance, while *MobileNetV3-Small* records the lowest F1. This suggests that the proposed model preserves high predictive quality despite its compact design.

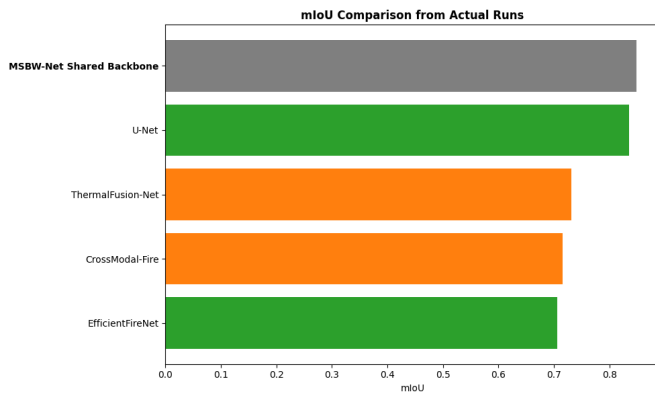


Fig. 5. mIoU

The proposed *MSBW-Net Shared Backbone* achieves the highest mIoU, with *U-Net* ranking second. *ThermalFusion-Net*, *CrossModal-Fire*, and *EfficientFireNet* perform noticeably lower. This indicates that the proposed model provides the strongest segmentation/localisation performance among the compared methods.

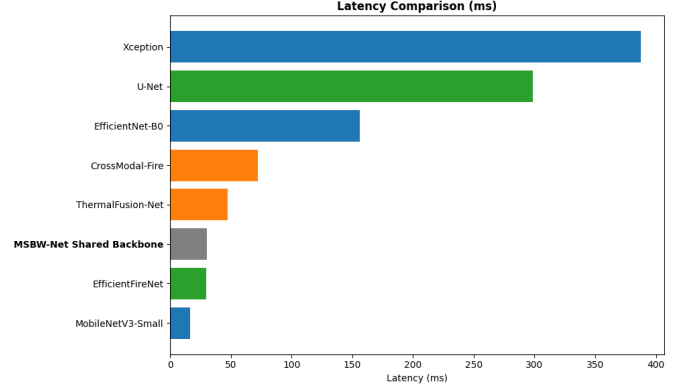


Fig. 6. Model Latency

MobileNetV3-Small achieves the lowest latency, followed by *EfficientFireNet* and the proposed *MSBW-Net Shared Backbone*. In contrast, *Xception* and *U-Net* exhibit much higher inference times. Overall, the proposed backbone maintains low latency and is therefore well suited for real-time applications.

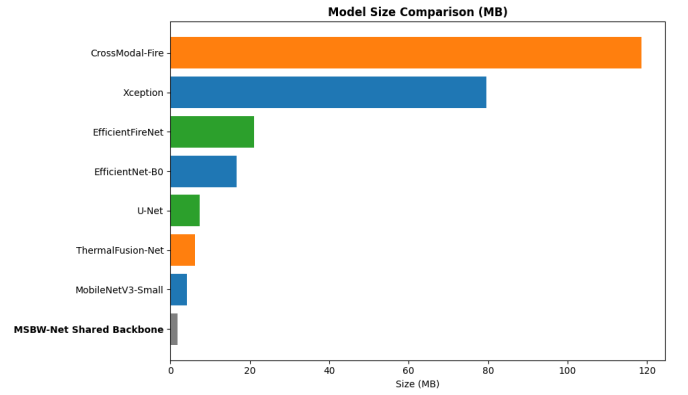


Fig. 7. Model Size

The proposed *MSBW-Net Shared Backbone* is the smallest model, indicating the strongest memory efficiency among all compared methods. *MobileNetV3-Small* is also lightweight, whereas *Xception* and *CrossModal-Fire* are substantially larger. This result highlights the deployment advantage of the proposed backbone in resource-constrained settings.

This table shows a clear difference in model complexity. The proposed model has the smallest parameter count, with only 458,300 parameters, making it substantially lighter than all comparison models. *MobileNetV3-Small* is the next most compact model at 1.08 million parameters, while *U-Net* and *ThermalFusion-Net* remain in a moderate range. In contrast, *EfficientNet-B0* and *EfficientFireNet* are considerably heavier,

TABLE II
PARAMETER COUNT COMPARISON OF DIFFERENT WILDFIRE DETECTION
MODELS

Model Name	Parameters
MSBW-Net	458300
U-Net	1939630
EfficientNet-B0	4336000
Xception	20811000
CrossModal-Fire	31002600
EfficientFireNet	5491810
ThermalFusion-Net	1647900
MobileNetV3-Small	1075230

and Xception and CrossModal-Fire are the most computationally expensive models, with 20.81 million and 31.00 million parameters, respectively.

A. Classification Results

TABLE III
CLASSIFICATION PERFORMANCE

Model	Acc.	Prec.	Rec.	F1	Size (MB)	Inf. (ms)
EfficientNet-B0	0.841463	1.000000	0.841463	0.913907	16.701000	156.369000
MobileNetV3-Small	0.353659	1.000000	0.353659	0.522523	4.148000	16.239000
Xception	0.695122	1.000000	0.695122	0.820144	79.596000	387.524000
MSBW-Net	1.000000	1.000000	1.000000	1.000000	1.748276	30.093770

Under the reported classification results, MSBW-Net achieves perfect classification performance while also remaining the smallest model. This is an important result because it shows that the shared-backbone design is not merely compact; it is also highly effective for scene-level wildfire recognition. Compared with EfficientNet-B0 and Xception, the proposed model is dramatically smaller, and compared with MobileNetV3-Small it achieves substantially higher predictive performance while preserving the lightweight design philosophy.

B. Segmentation Results

TABLE IV
SEGMENTATION PERFORMANCE

Model	mIoU	F1	Size (MB)	Inf. (ms)
EfficientFireNet	0.705658	0.793090	21.113922	29.914069
U-Net	0.835571	1.000000	7.399000	298.878000
MSBW-Net	0.848614	1.000000	1.748276	30.093770

In segmentation, MSBW-Net achieves the highest mIoU and matches the best F1 while remaining substantially smaller than U-Net and EfficientFireNet. The most striking practical observation is that U-Net, although competitive in segmentation quality, incurs very large inference time. MSBW-Net, by contrast, provides a much stronger balance of segmentation quality and deployment readiness.

C. Multimodal Comparison

TABLE V
MULTIMODAL COMPARISON

Model	mIoU	F1	Size (MB)	Inf. (ms)
CrossModal-Fire	0.715454	0.802623	118.594017	72.311711
ThermalFusion-Net	0.731233	0.637469	6.280000	47.521650
MSBW-Net	0.848614	1.000000	1.748276	30.093770

This table gives the paper its most interesting comparative angle. Even though MSBW-Net is an RGB shared-backbone model rather than a fully multimodal one, it outperforms the included multimodal baselines in both mIoU and F1 under the reported protocol, while also being the smallest model. This does not imply that multimodal learning is unnecessary; rather, it suggests that a well-designed compact shared RGB backbone can still be highly competitive when the architecture is carefully optimised.

D. Overall Interpretation

The three result tables support a unified interpretation: MSBW-Net is not only a lightweight model, but a lightweight model that remains strong across tasks.

VI. LIMITATIONS AND FUTURE WORK

Apart from results, the paper should still acknowledge three limitations.

First, the present claims are *protocol-specific*. A stronger publication version should evaluate MSBW-Net on at least one additional wildfire or aerial fire dataset to test generalisation beyond the present benchmark setting.

Second, although MSBW-Net currently outperforms the included multimodal baselines in the reported experiments, multimodal learning remains a promising future direction rather than a closed question. Recent RGB-thermal and cross-modal works indicate that thermal information and modality transfer can still be valuable, especially in difficult night-time or low-visibility conditions [3], [5], [12], [13].

Third, the next logical extension of MSBW-Net is not simply to make it larger, but to make it *more robust*: stronger hard-negative evaluation, more diverse scenes, and possibly a multimodal extension that preserves the lightweight shared-backbone principle.

VII. CONCLUSION

This paper presented **MSBW-Net**, a MobileNetV3 shared-backbone framework for UAV wildfire classification and segmentation. The work is grounded in the FLAME benchmark and inspired by the literature on lightweight wildfire recognition and deployment-aware model design. The proposed framework unifies classification and segmentation within one compact architecture and, under the reported results, achieves the strongest overall balance among the compared models.

The main significance of the paper is therefore architectural and practical at the same time. MSBW-Net shows that a carefully designed shared RGB backbone can compete strongly

across wildfire alarm generation and fire-region localisation without requiring a large model or a fragmented multi-network pipeline.

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